
SSM Optimization

Weekly Meeting

24/04/26

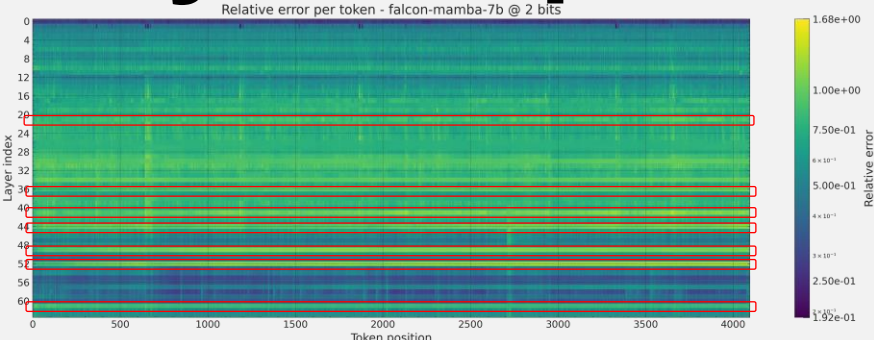
Experiment setup

- Run inference over a fixed sequence and examine the state divergence (FP vs Q) of all SSM blocks of the network
- *Input*: Wikitext2 batches, and custom prompts
- *Models*: Falcon-Mamba-7b, Mamba-2.8, Mamba-1.4b

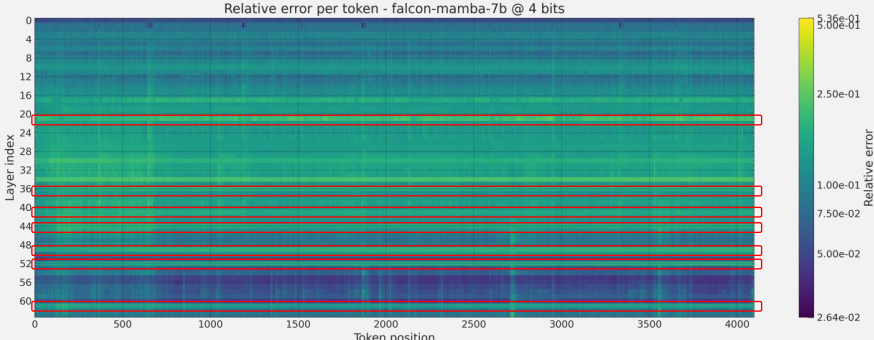
Layer vs position: bit-width

Falcon-Mamba-7b

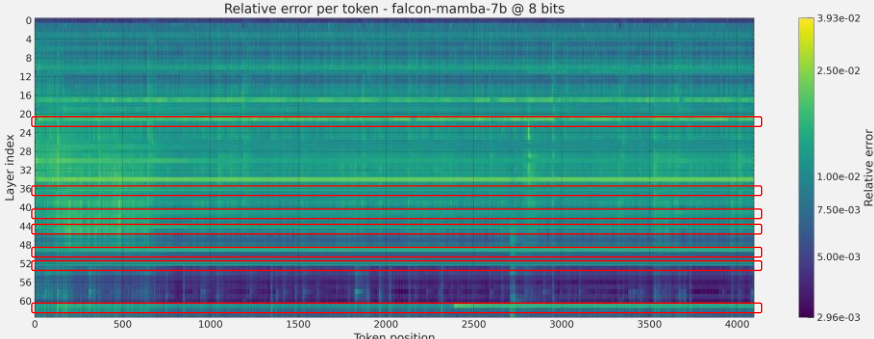
2 bits



4 bits

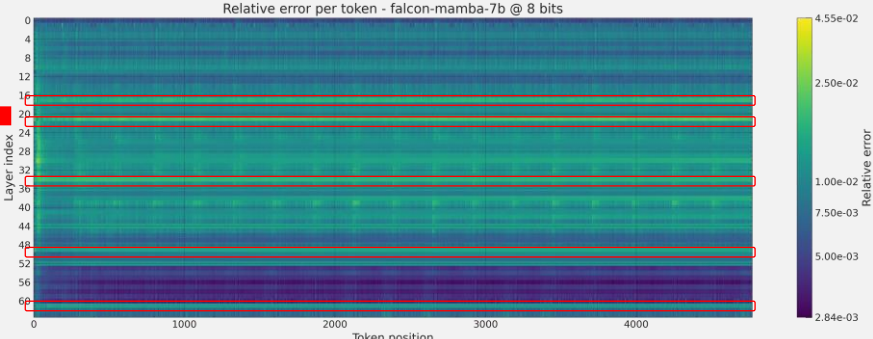
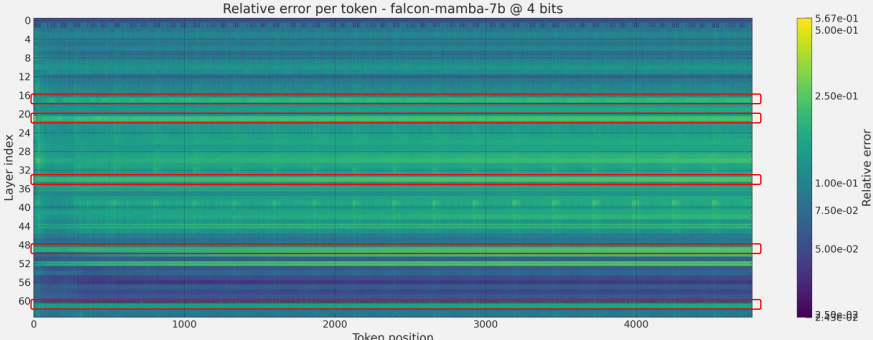


8 bits



More pronounced rel. error than at other bit-widths

Most patterns are consistent across bit-widths



Layer vs position: bit-width

Mamba-2.8b

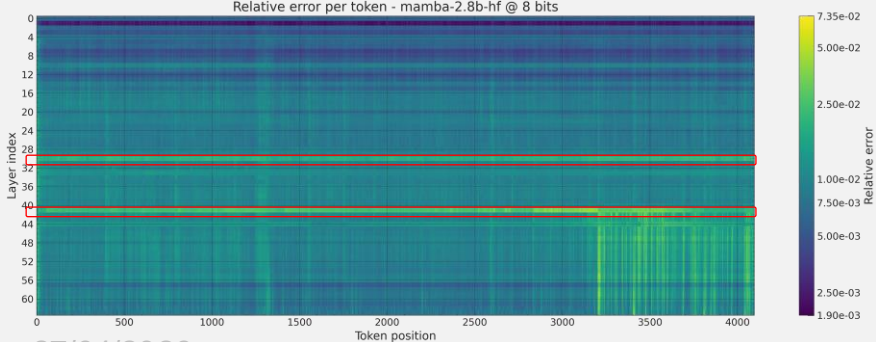
2 bits



4 bits



8 bits



↓
Behaviour seems to be not model-specific



Layer vs position: bit-width

Mamba-1.4b

2 bits



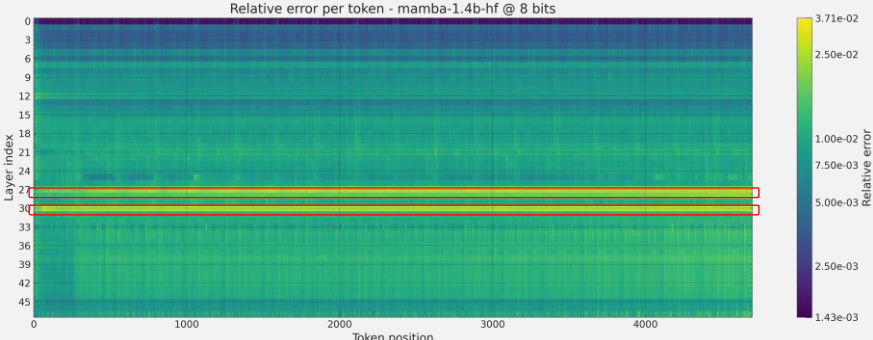
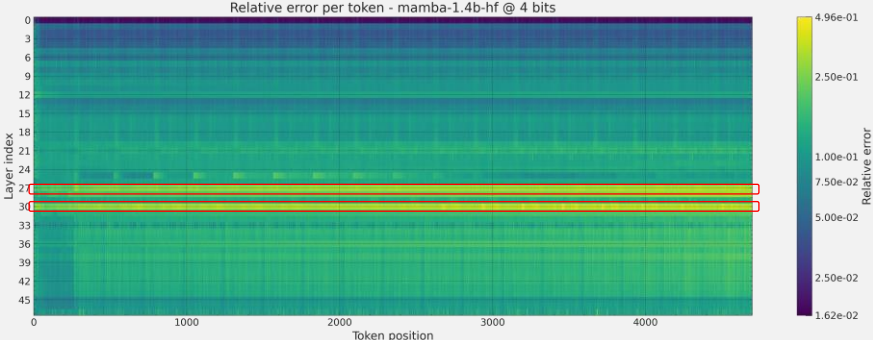
4 bits



8 bits



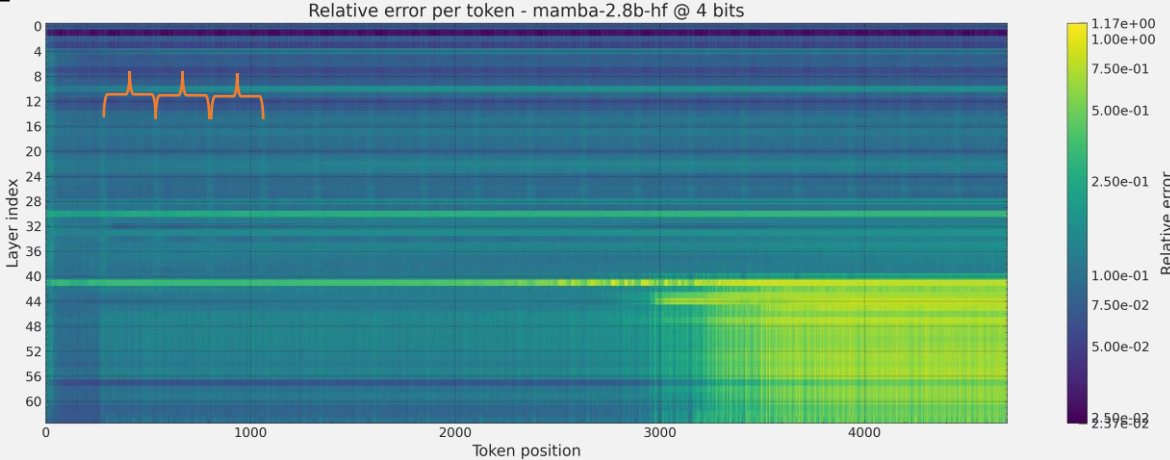
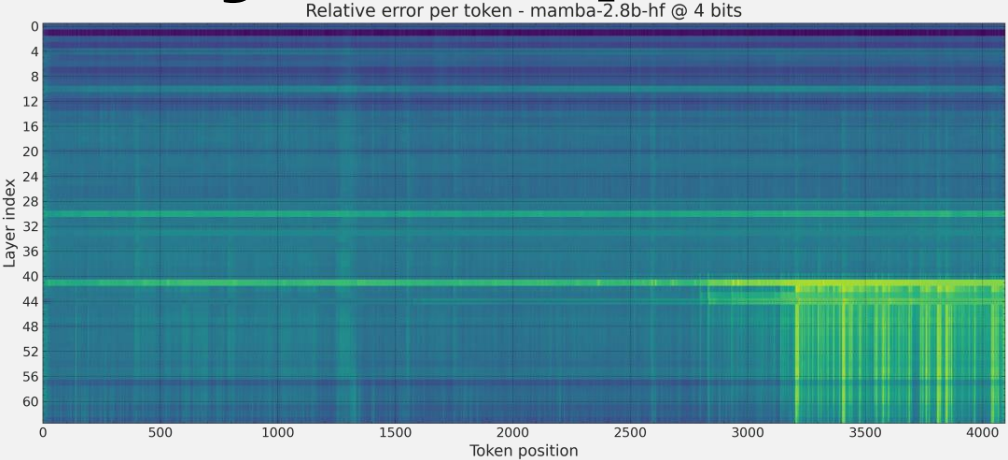
↓
Behaviour seems to be not model-specific



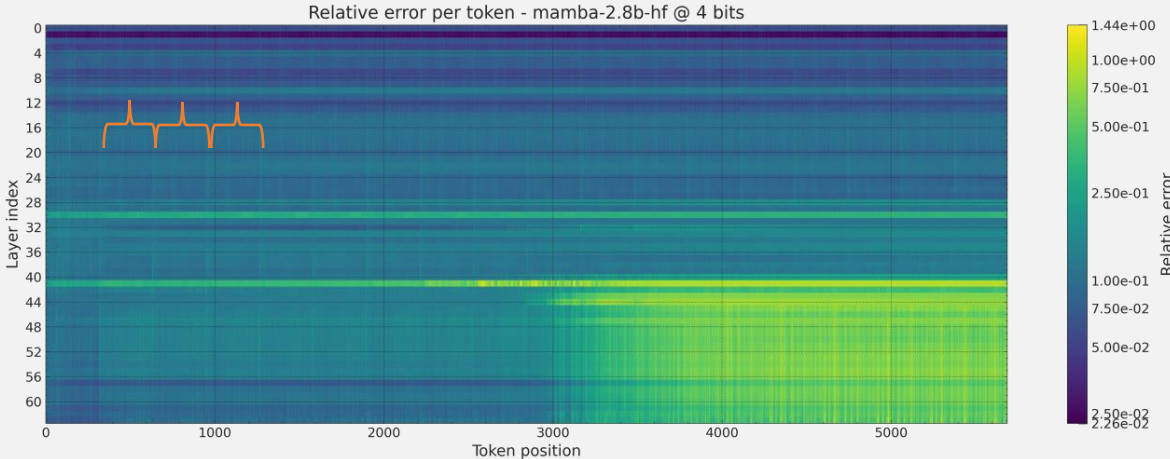
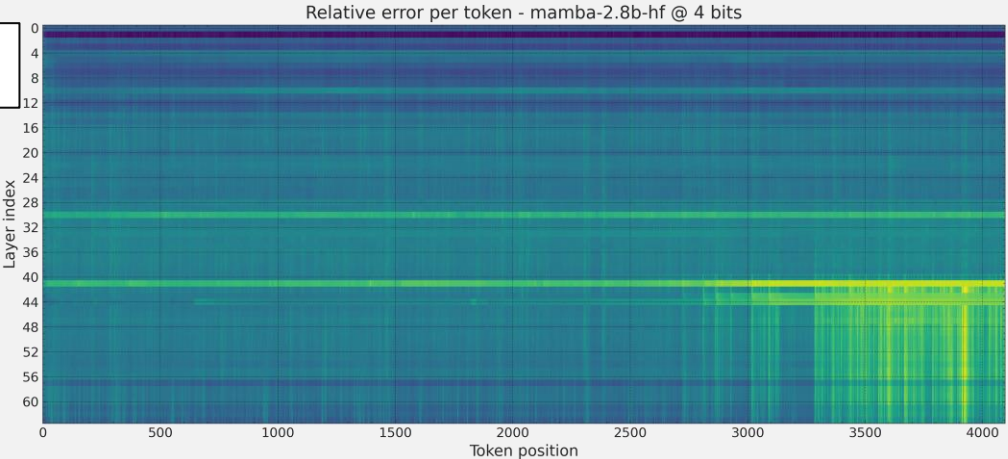
Layer vs position: input

Mamba-2.8b @ 4 bits SINQ

B0



B50



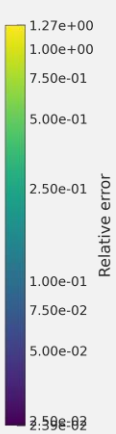
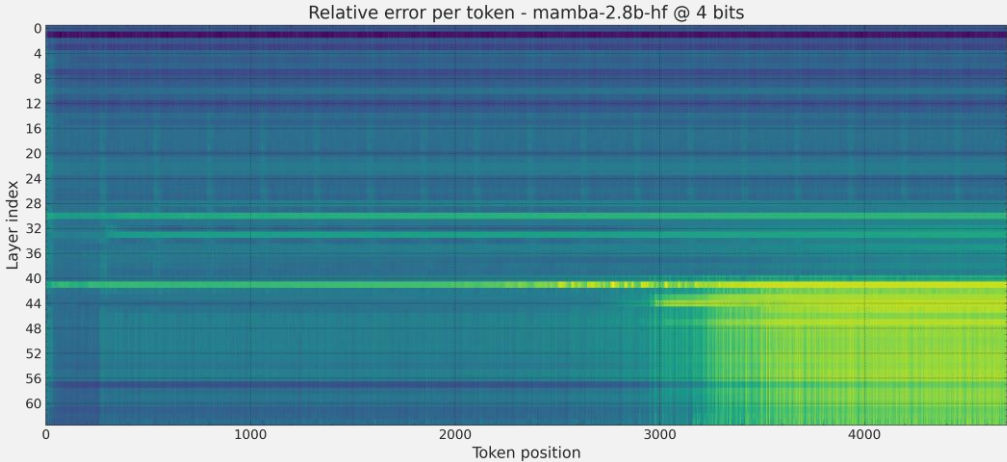
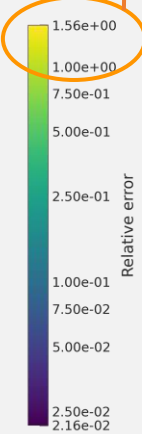
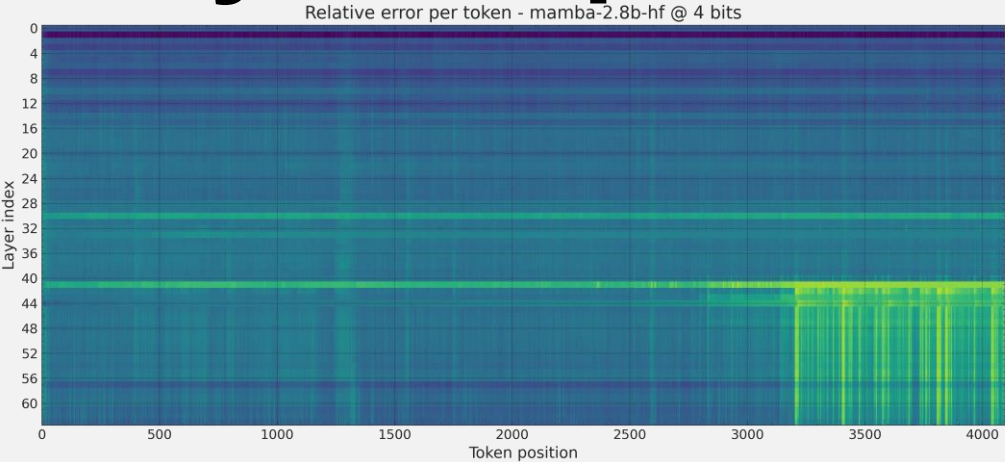
State drift

- Slightly higher relative errors
- Very very similar error trend

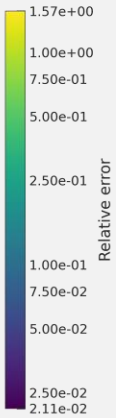
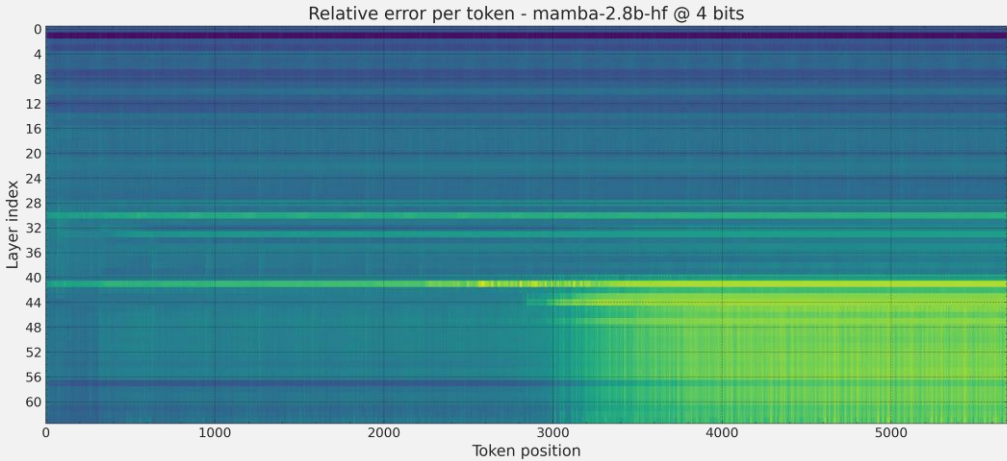
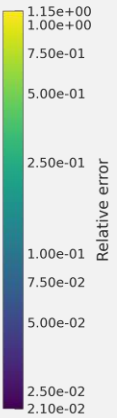
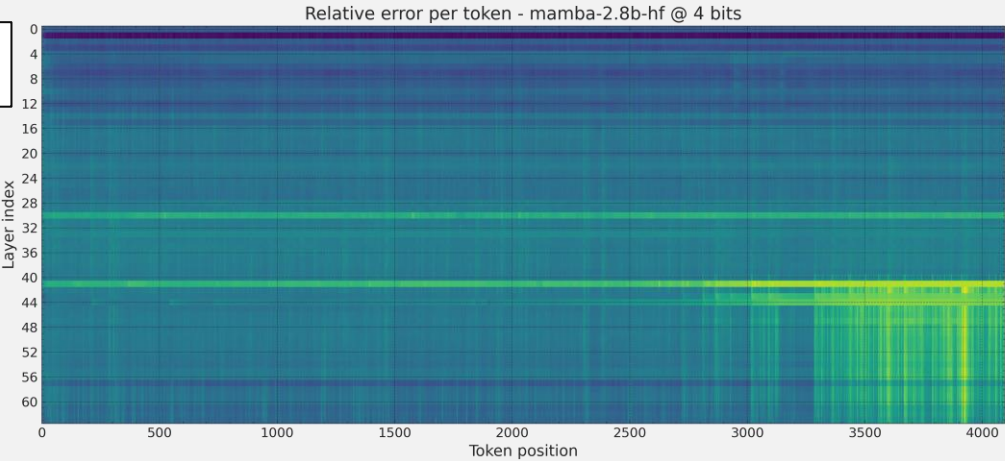
Layer vs position: RTN

Mamba-2.8b @ 4 bits RTN

B0



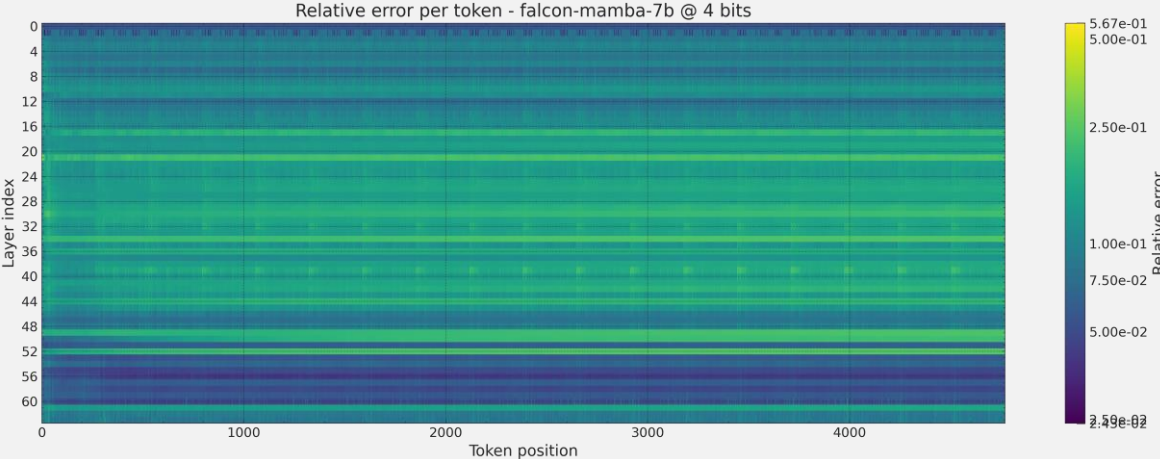
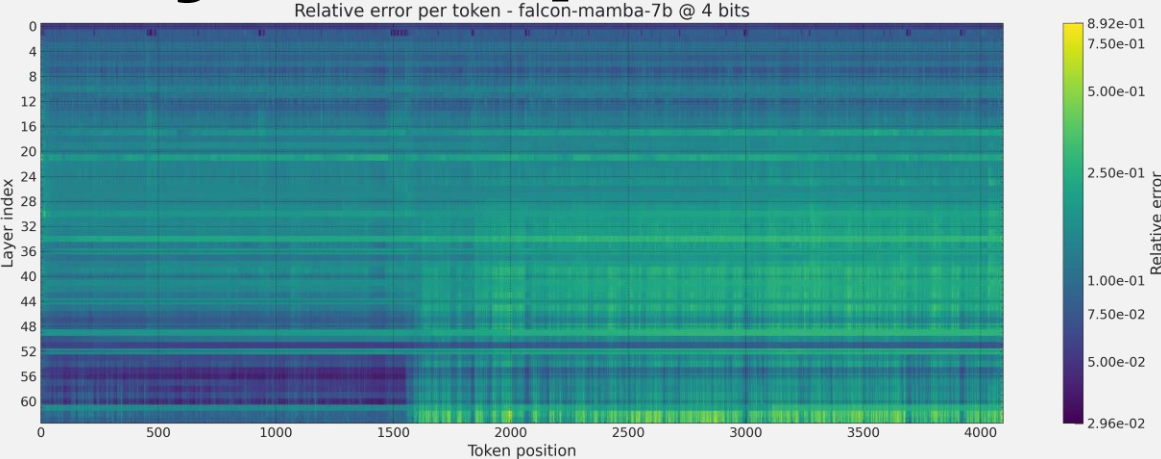
B50



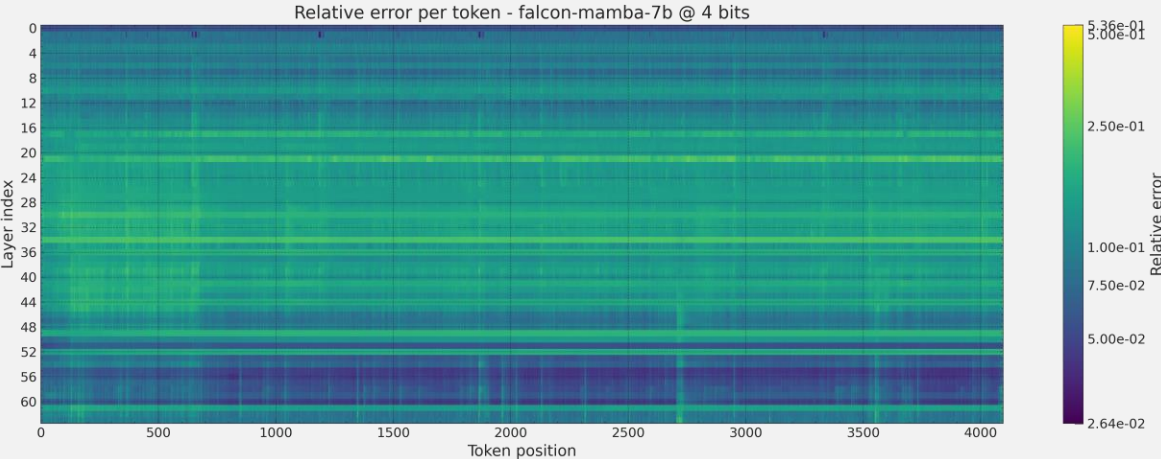
Layer vs position: SINQ

Falcon-Mamba-7b@ 4 bits SINQ

B0



B50



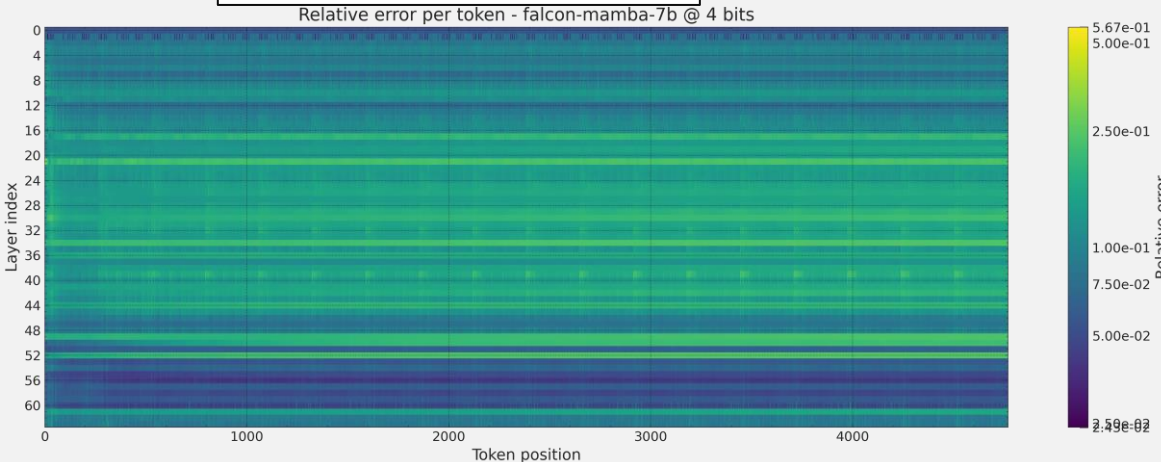
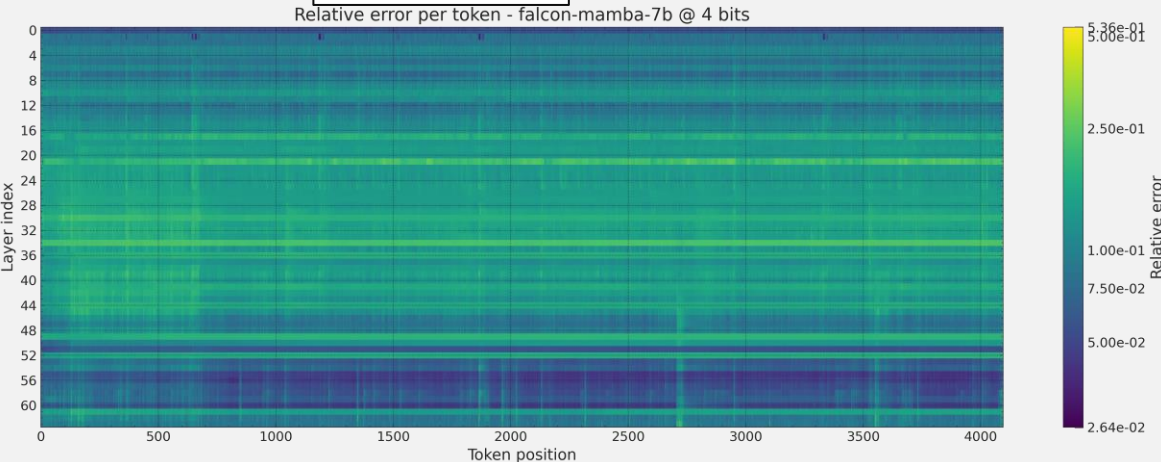
Layer vs position: SINQ

Falcon-Mamba-7b@ 4 bits SINQ

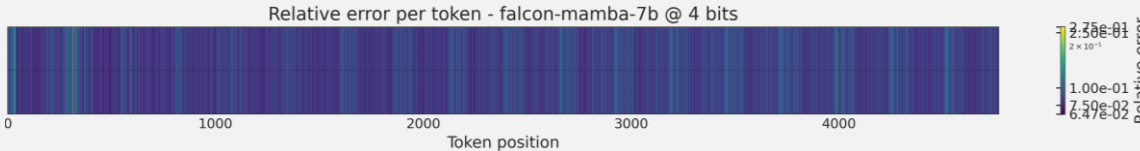
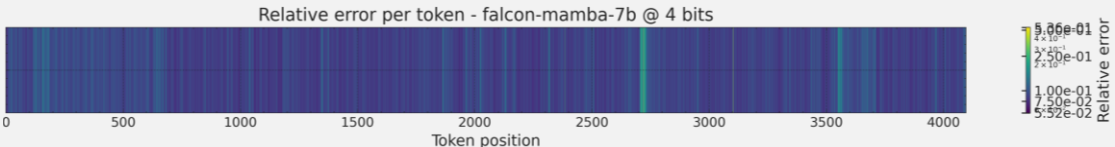
Wikitext2

Repeated patterns

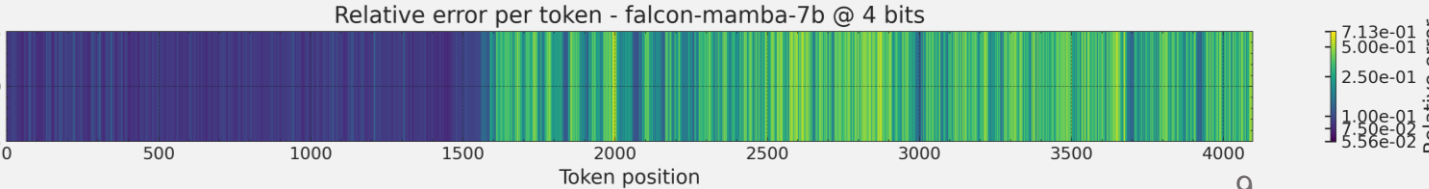
B50



Last layer



As a reference, B0 of Wikitext2:



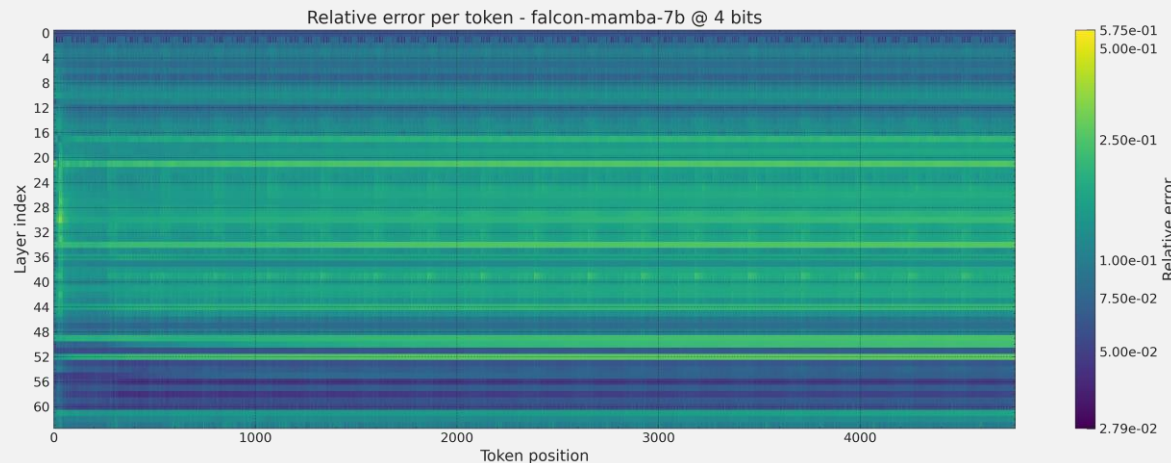
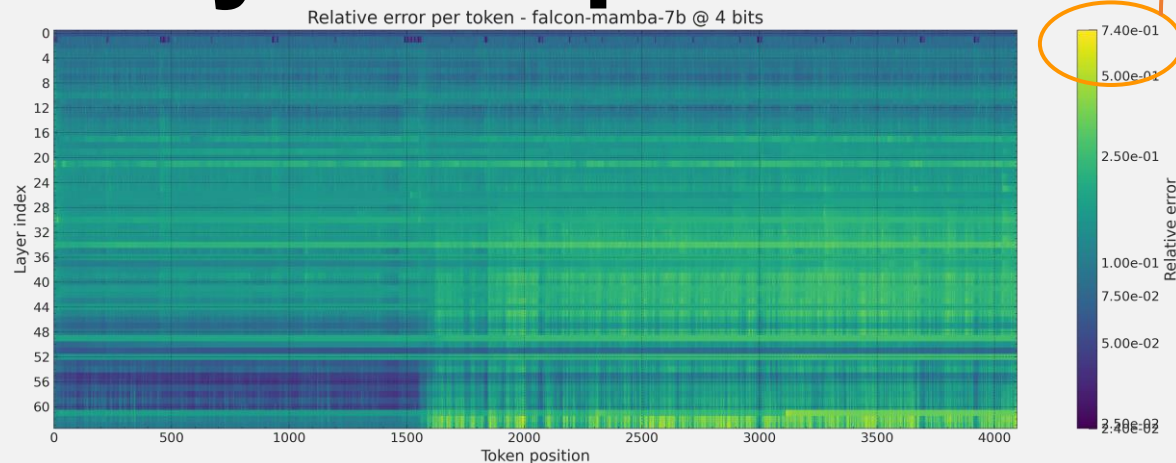
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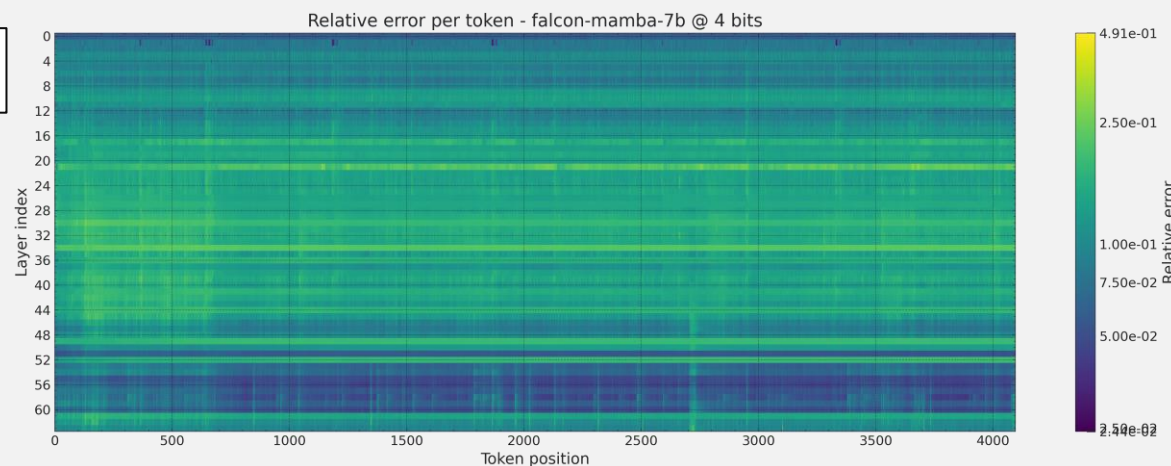
Layer vs position: RTN

Falcon-Mamba-7b@ 4 bits RTN

B0



B50



Conclusions

- Mamba-2.8b exhibits drift over time regardless of the input properties
- Falcon-Mamba-7b's behaviour depends a bit more on the input sequence
- SING vs RTN → similar output patterns
- Some layers are the main contributors to drift, regardless of the input/model bit-width

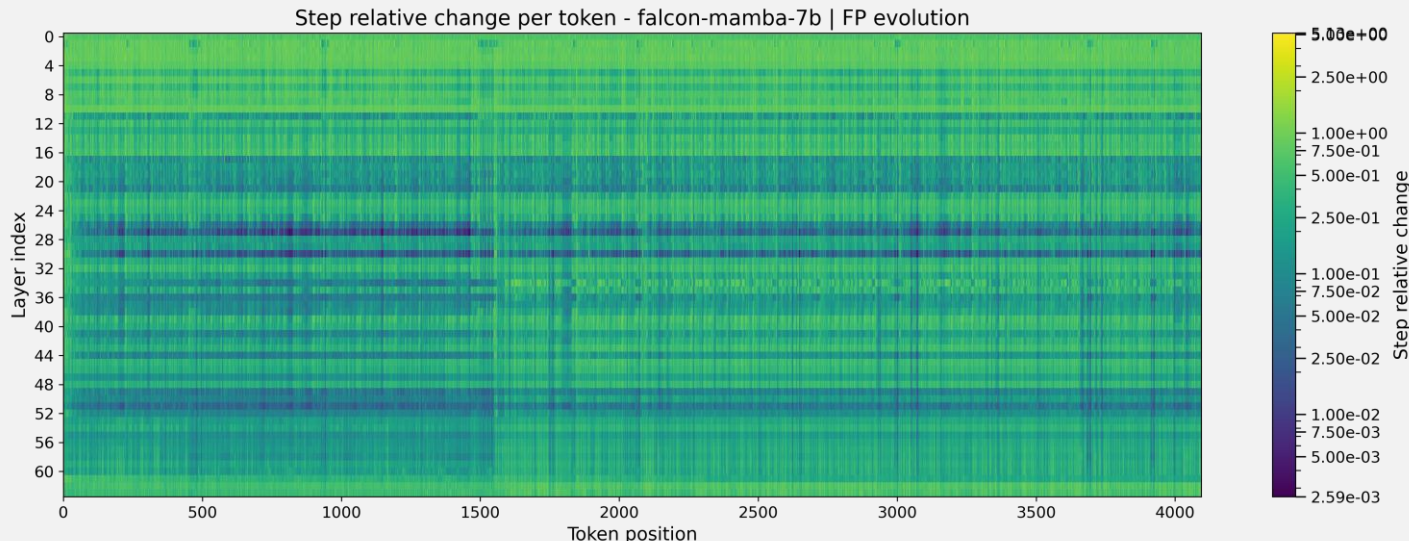
Experiment setup

- Run inference over a fixed sequence and examine the state evolution of all SSM blocks of the network
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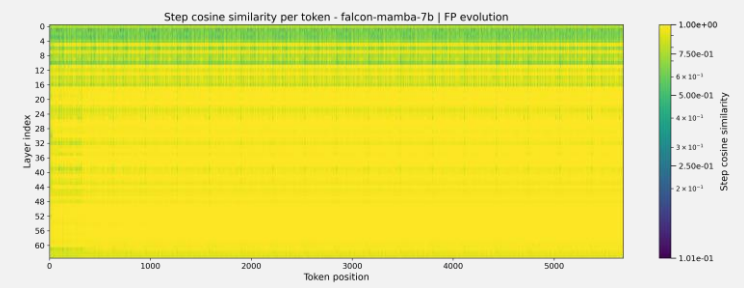
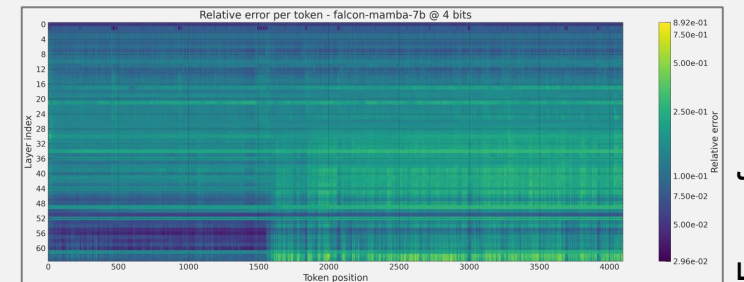
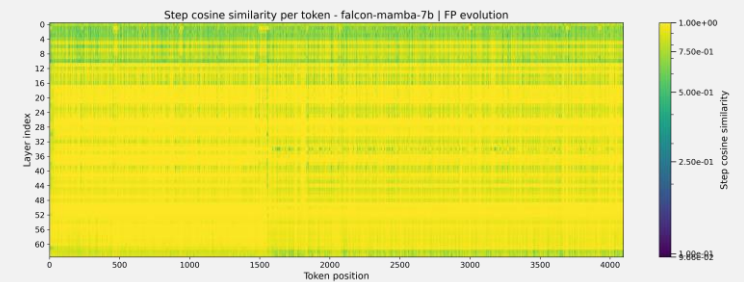
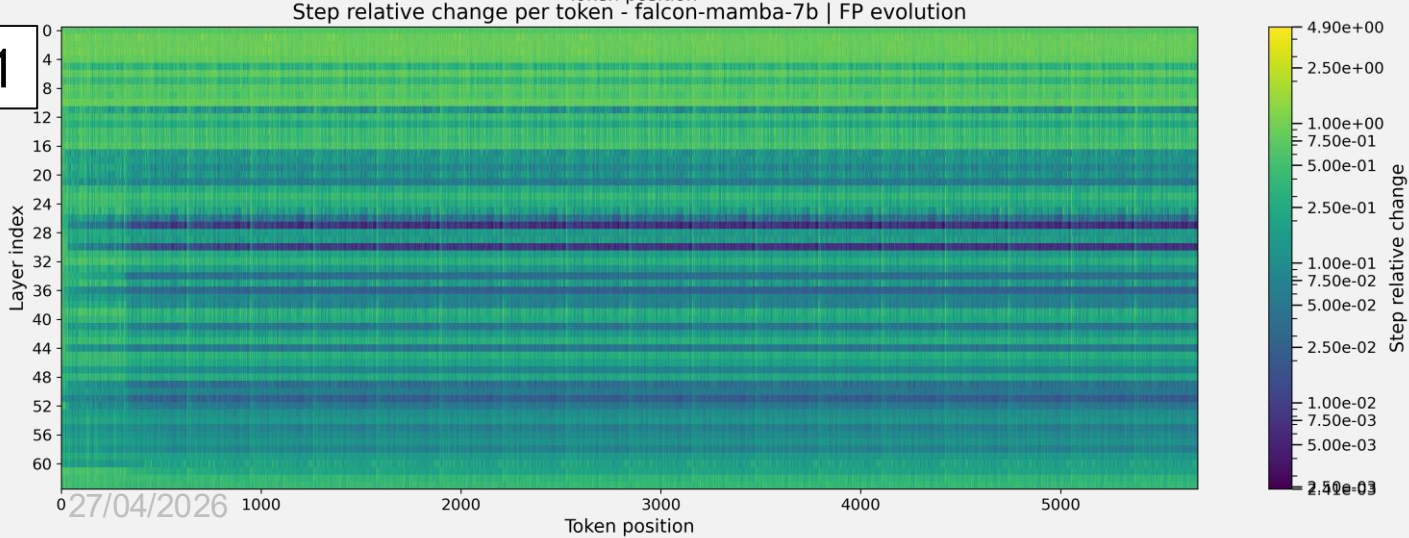
FP State evolution

Falcon-Mamba-7b

B0



Rep1

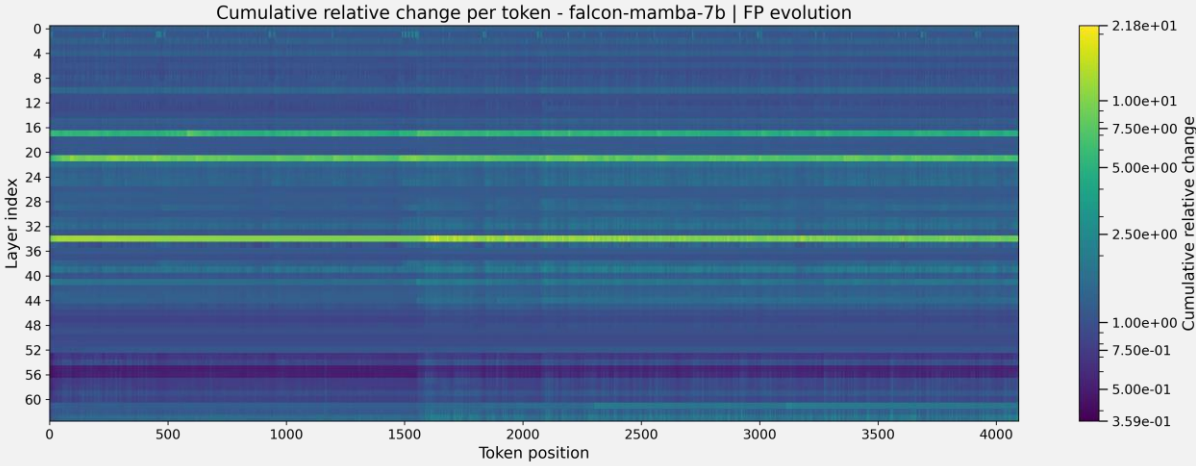
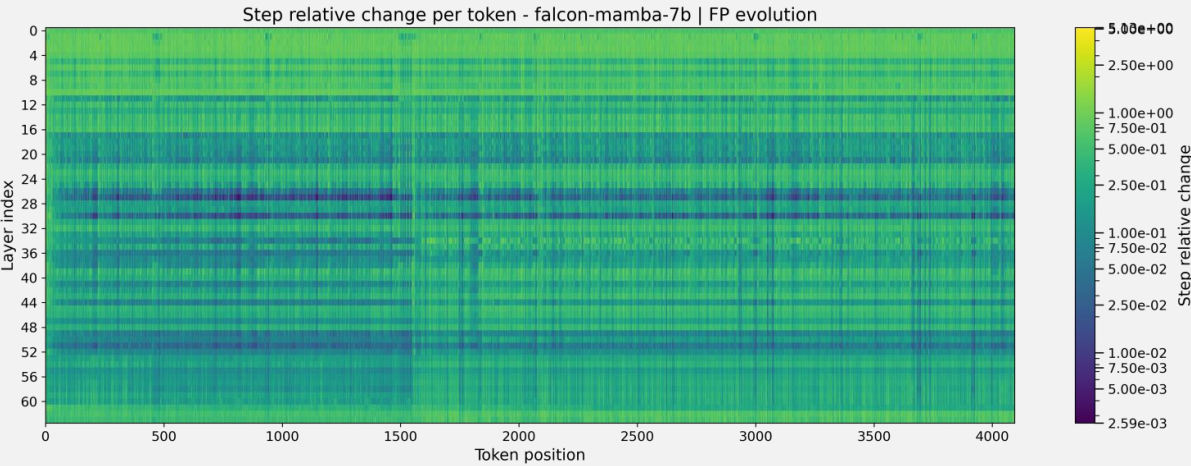


For reference:
quantization error

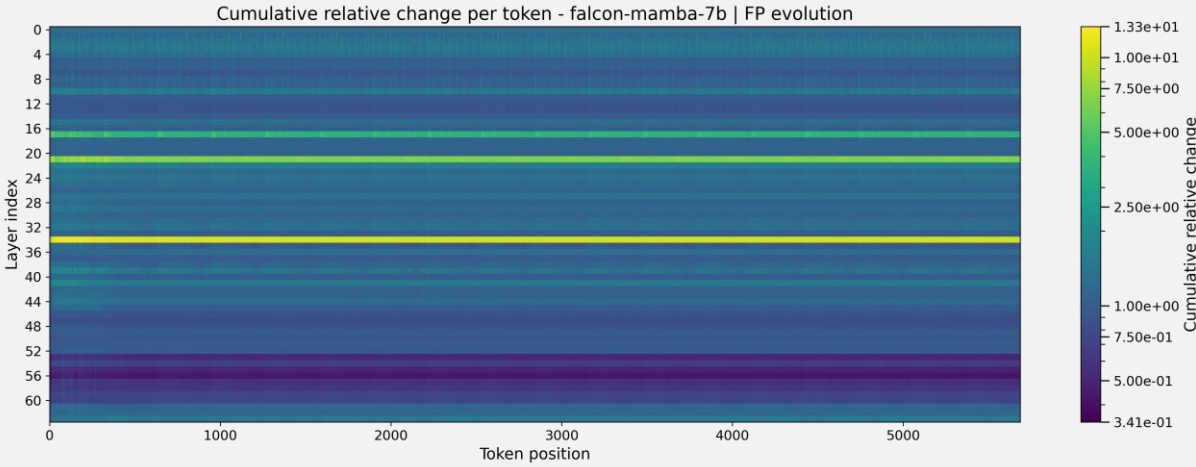
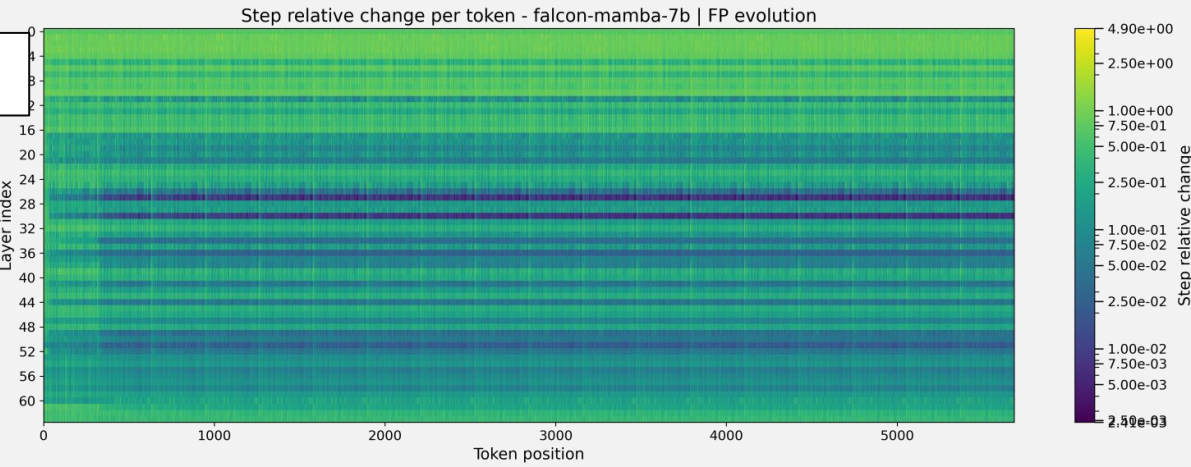
FP State evolution

Falcon-Mamba-7b

B0



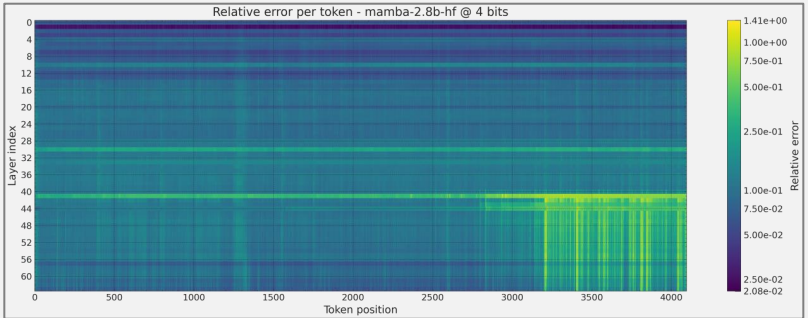
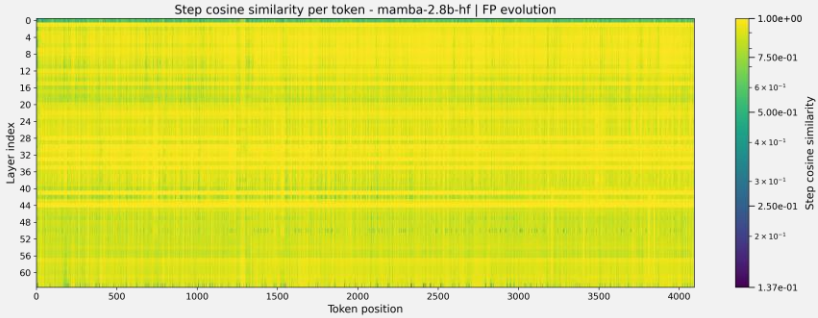
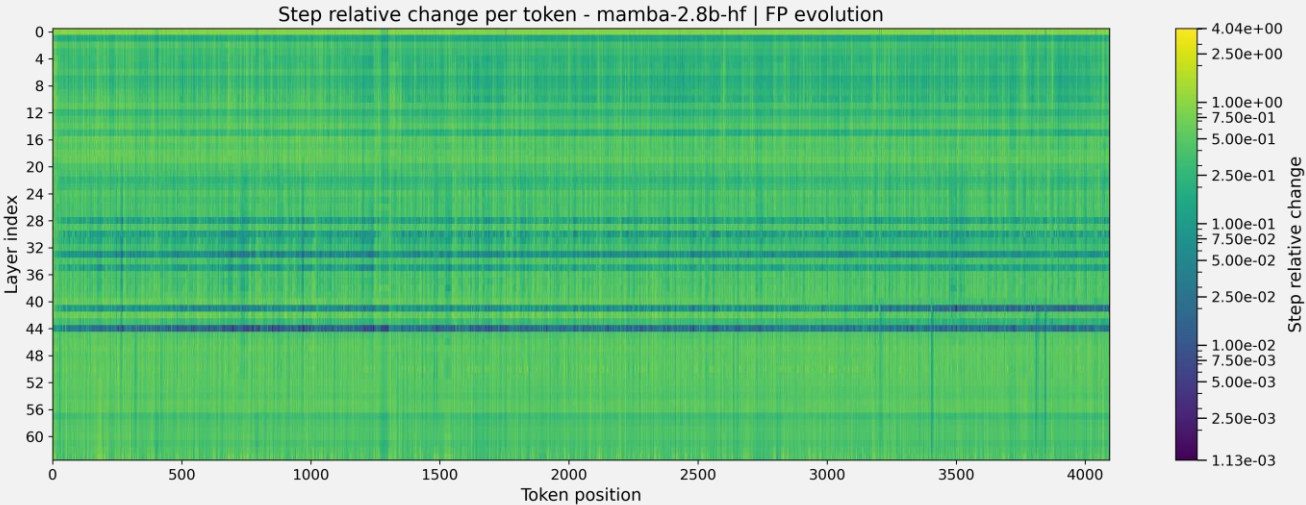
Rep1



FP State evolution

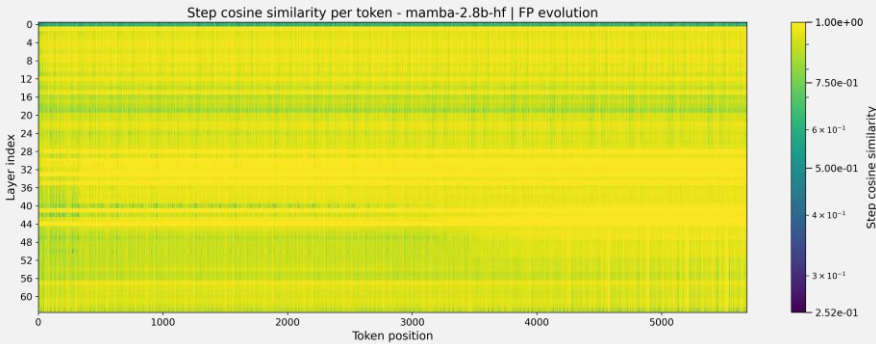
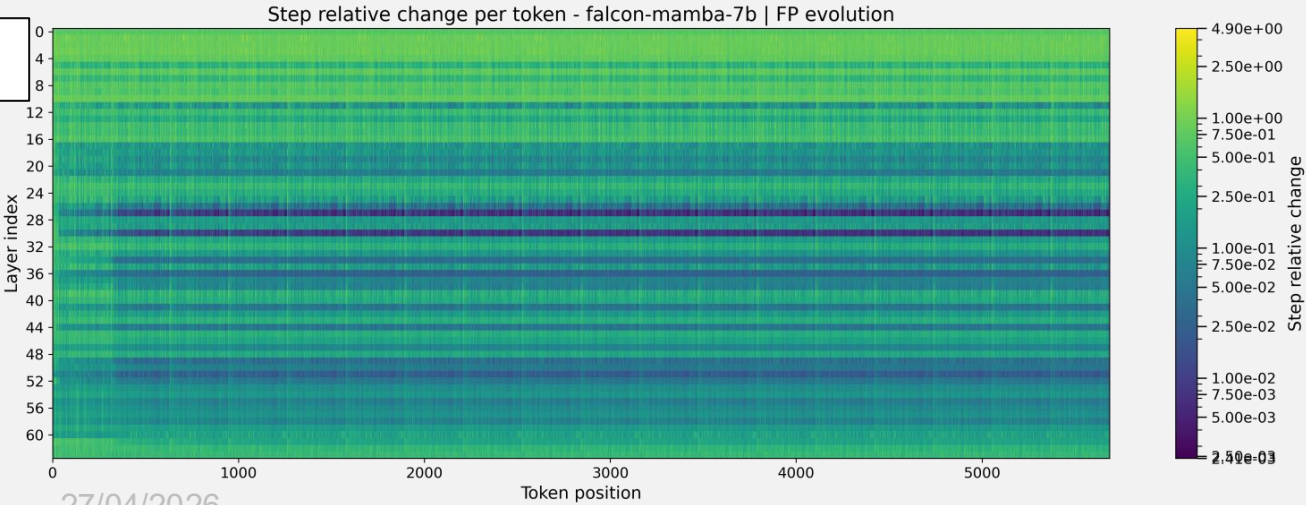
Mamba-2.8b

B0



For reference:
quantization error

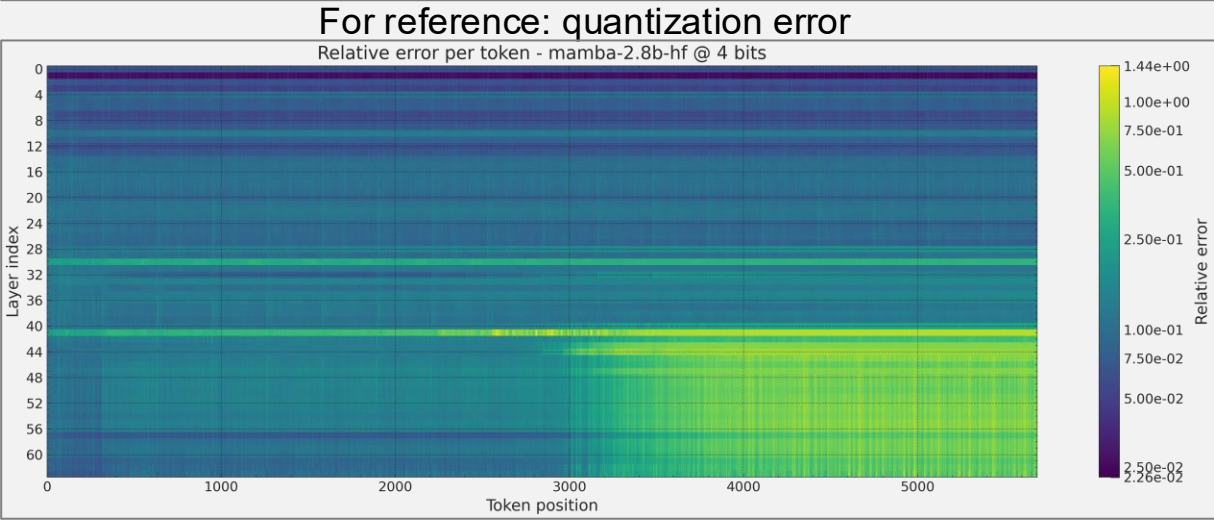
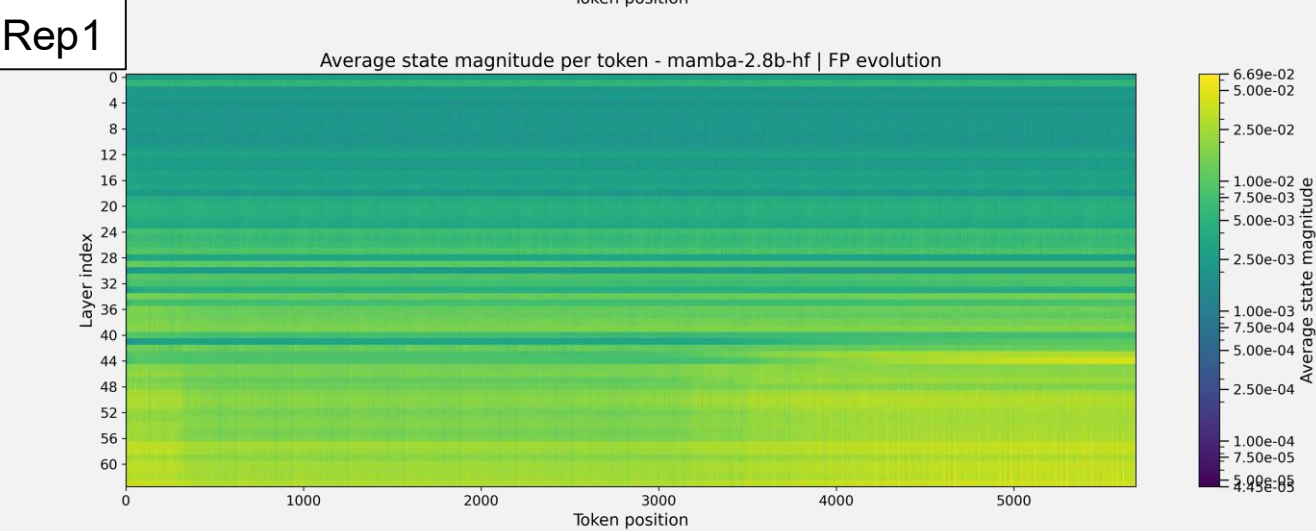
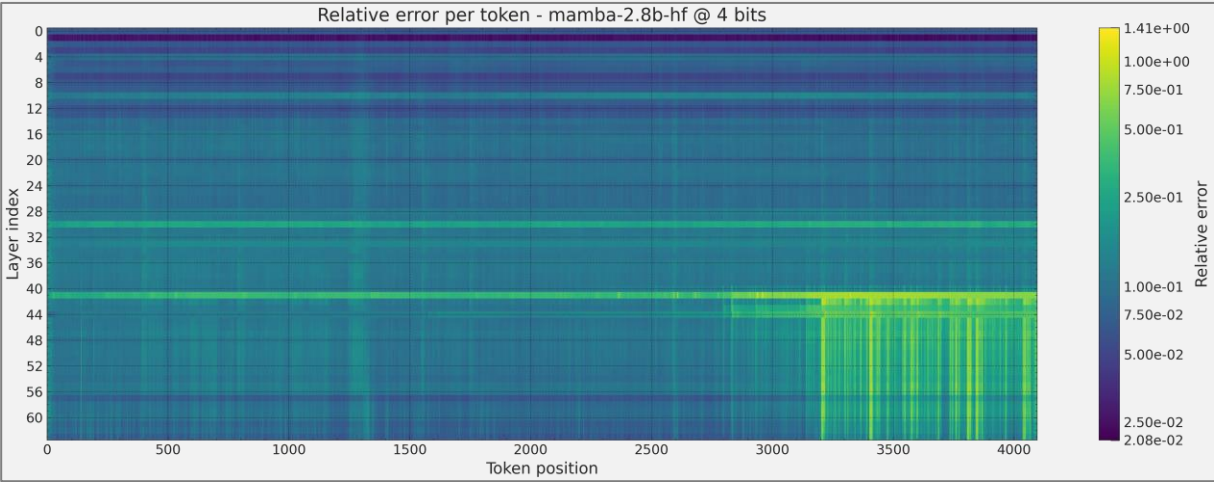
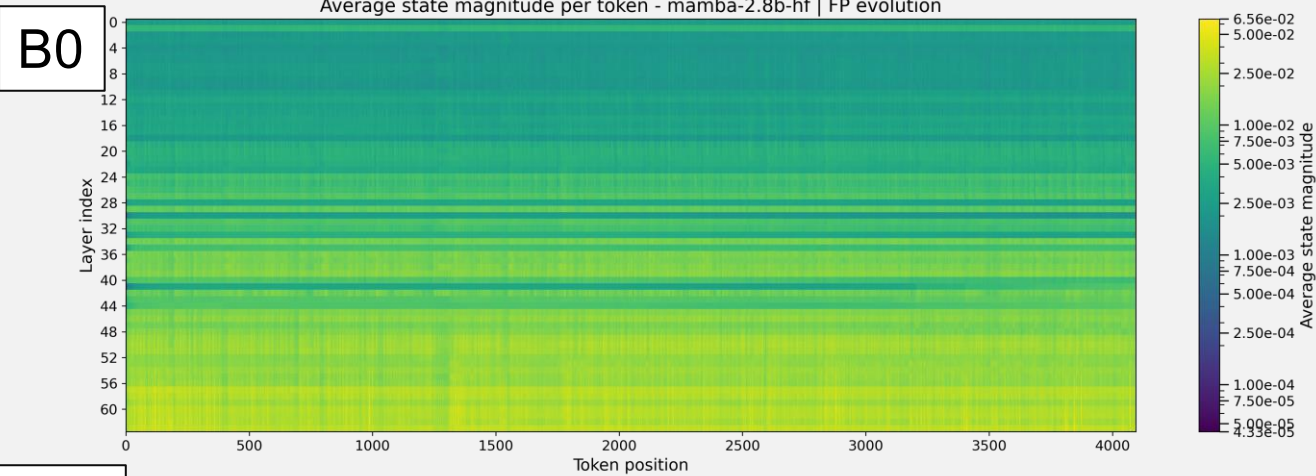
Rep1



FP State evolution

Average state magnitude

Mamba-2.8b

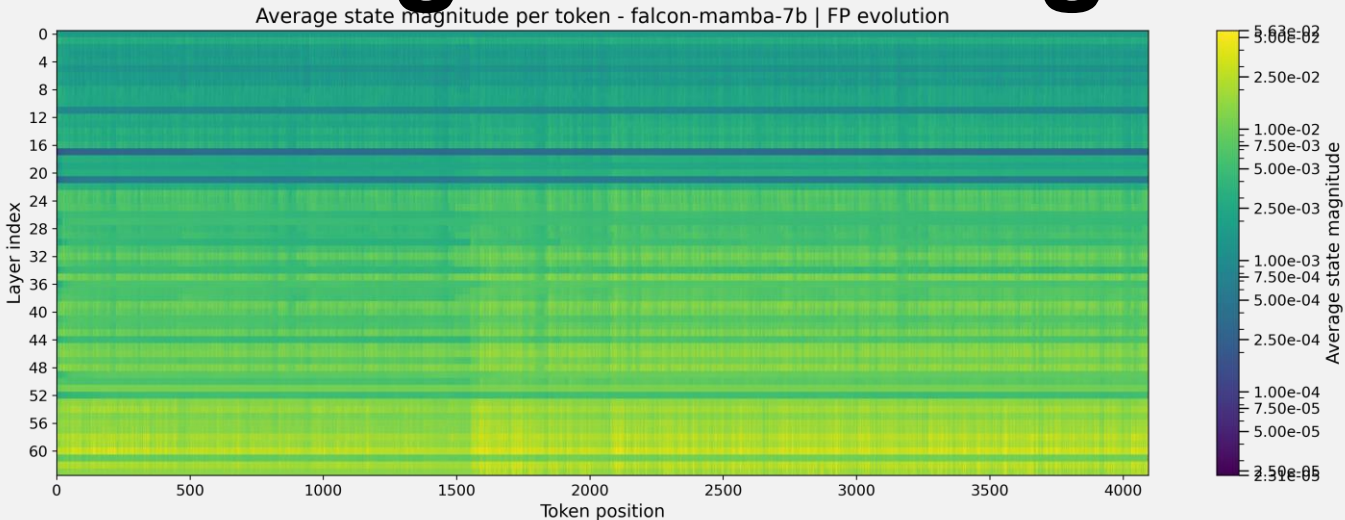


FP State evolution

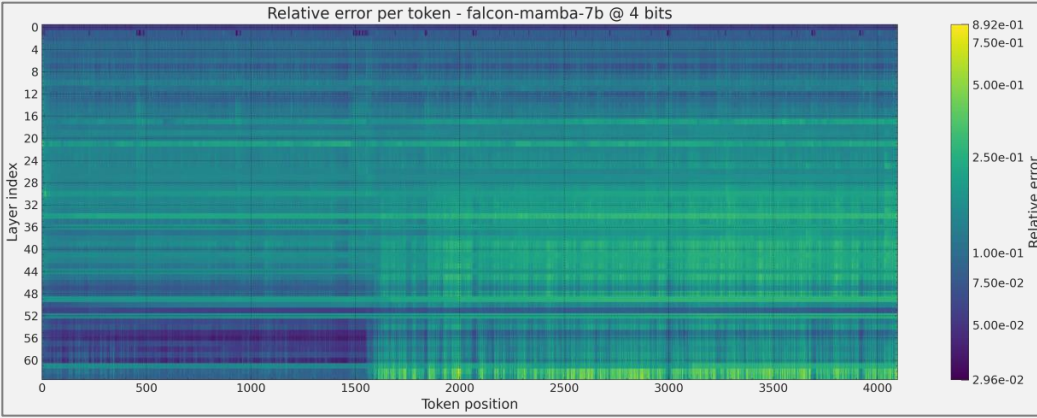
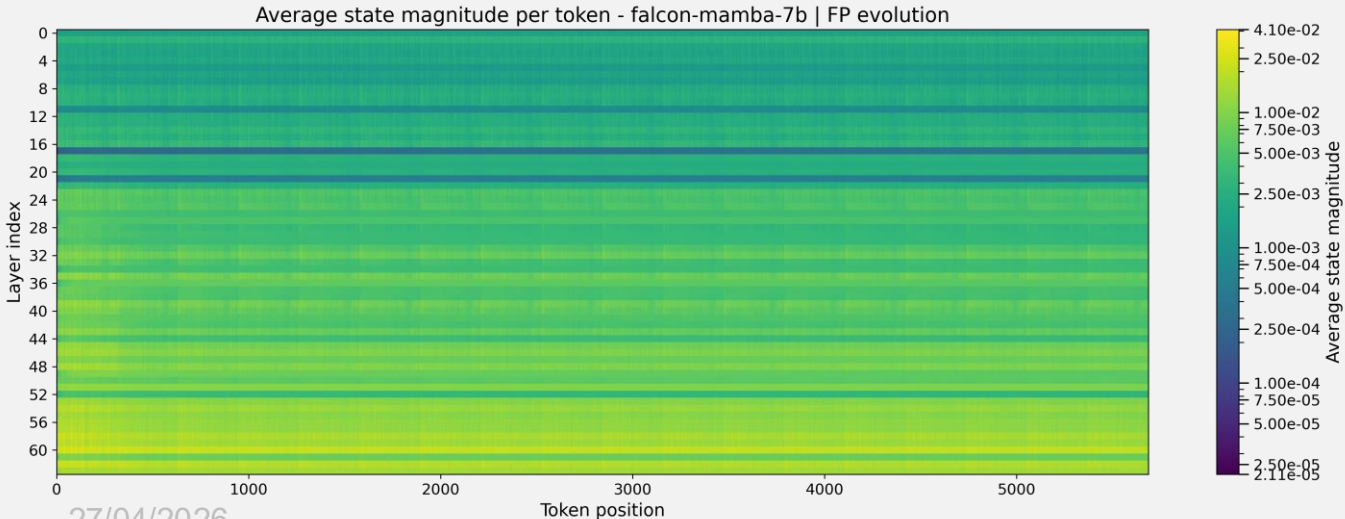
Average state magnitude

Falcon-Mamba-7b

B0



Rep1



For reference: quantization error

Next steps

- Continue working on activations quantization
- Hybrid architectures

+ Other paper (WIP)